A34: checks and closures for the clean Category A ledger

For: Arvin Hampton · 2026-10-22, America/Chicago (CDT).

Ground rules.

- New files only, in `SF/A34_ledger_checks/`.
- CORE_FREEZE is respected: H_s is not redefined, and no certifier code was touched or run.
- The s31 mm sector was not touched.
- All jobs ran under `nice` with OMP_NUM_THREADS=1 and `ulimit -v` ≤ 2.5 GB. Peak resident memory was about 0.7 GB.
- Nothing was pushed or sent. No TPC/RH claims are made.

Categories. Cat A = proved, or exact relative to named hypotheses. **Cat B** = numerical or interpretive. The two are kept strictly apart.

`SF/` = `/workspace/quantum-comp/reports/s_family_paths/`. Entry numbers refer to `LEDGER_CatA_clean` (v1).

Summary

Item	Outcome
(1) Entry 14, closed forms for d_cusp	**Fixed.** The odd-s statements are verified. The even-s statements are only conditional, and one step has a gap (§1.1).
(1) Entry 17, Gauss-sum vanishing	**Fixed.** The statement had a transcription error: it read "2 ‖ m" where the correct condition is "2 | m". The full proof is verified, and the exact values of |G| were added (§1.2).
(1) Entry 23, Hecke commutation	**Verified** under the hypothesis of Γ(N)-invariance (TB4). The proof is written out (§1.3).
(1) Entry 24, sector lemma	**Fixed.** The proof needs dim V = dim W, which is proved only for odd s ≤ 51, so the scope is restricted to that range (§1.4).
(2) Certificate 10	**Fixed citation.** The Y-certificate is `SF/A8c_Spairing_even/scripts/pairing_on_W.py`, not `A8_Spairing_even/modp_trace.py`. Re-run for s = 8, 16, 18, 24, 26: identical results. `global_trE.py` was also re-run: identical results.
(2) Certificate 56	**Verified** (`galois_dilation_fast.py`). Re-run at s = 8 and 10: identical results.
(2) Certificate 57	**Fixed.** The script `classgroups_A5.py` uses PARI `quadclassunit`, so I replaced it with

Item	Outcome
	an unconditional reduction-cycle computation (`A34/scripts/classno_exact.py`). The values agree. **The claim "the Cl(4n_s) predictor fails at s = 25" was wrong**: the predictor gives 3, and dim S(ρ_25) = 3. The correct exact failures are listed in §2.3.
(2) Certificate 58	**Fixed citation.** The (187, 186, 187) multiplicities came from `scan.py`, which is floating point (Cat B). The entry now has a proof: entry 11 together with an exact dim E = 560 (§2.4).
(3) rev1 entry 15, distribution of r	**Closed** (Cat A). The CHECK marks are resolved. The mean upper bound follows from Nair–Tenenbaum (Acta Math. 180 (1998), Theorem 1), whose hypotheses I verified. The lower bound is elementary. The Gaussian refinement is not needed and is not included (§3.1).
(3) Friction-coupled PDE	**Closed for its spatially homogeneous reduction.** In that reduction there is no attracting solution of any kind, and the stated function is not a solution. **The full PDE is not well-formed**: the contraction for γFΔΦ is unspecified, so nothing can be proved about it (§3.2).
(3) Class-group predictor at s = 27, 29	**Failed to close.** dim S(ρ_s) ∈ [3, 4] at both s, and the predictor value is 3. Raising the Hecke lower bound with 8 primes leaves it at 3 (§3.3). Only the Cl(K_s) version fails exactly there.
(3) E_L ↔ u_L identification	**Not proved; left out.** It is an analytic identification of a cone-sum generating function with a Maass theta lift. None of the ledger entries depends on it (§3.4).
Found during checks	Entry 2 (s ≡ 4, 6 mod 8) **failed as stated**. A6 verified the even-s step only for s ≤ 38. It is replaced by an exact certificate for 12 ≤ s ≤ 46 (§2.5). As a result, the corollary "tr_C S = 0 for every s ≥ 5" is narrowed to the proved ranges.

1. Proof checks

1.1 Entry 14 (closed forms for d_cusp; `SF/A/REPORT.md` §1, `p1_formula_check.py`)

The setting is odd s with n = s(s−2), G(m) = ∏_{q‖m} (q+1)/2, dim V = s(s−1) and d_Eis = 3G(n).

- **Lemma (verified).** For odd m, G(m) ≤ (m+1)/2, with equality iff m = 1 or m is a prime power.
- *Proof.* If m = q is a prime power, G = (q+1)/2.
- For coprime odd a, b ≥ 3, we have (a+1)(b+1)/4 < (ab+1)/2. This is equivalent to (a−1)(b−1) > 0.
- If G(a) ≤ (a+1)/2 and G(b) ≤ (b+1)/2, then G(ab) = G(a)G(b) ≤ (a+1)(b+1)/4 < (ab+1)/2.
- Induction on the number of prime-power factors gives the lemma, with strict inequality whenever there are two or more factors.
- **Odd s lower bound (verified).**
- Since gcd(s, s−2) = 1, G(n) = G(s)G(s−2) ≤ (s+1)(s−1)/4.

- So d_cusp = s(s−1) − 3G(n) ≥ s(s−1) − 3(s²−1)/4 = (s−1)(s−3)/4 > 0 for s ≥ 5.
- Equality holds iff s and s−2 are both prime powers.
- **Odd s parity (verified).**
- s(s−1) is even.
- One of s, s−2 is ≡ 3 mod 4. An integer ≡ 3 mod 4 has a prime-power factor q ‖ it with q ≡ 3 mod 4, because a product of factors ≡ 1 mod 4 is ≡ 1 mod 4.
- For such q, (q+1)/2 is even, so G(n) is even and d_Eis is even.
- **Even s: d_cusp = 0 ⇔ s ∈ {6, 8, 10}, and parity (NOT verified unconditionally).**
- The proof assumes $d_2(k, v) \le 2^{k-1} + 6$, which is proved only for k ≤ 6.
- Step 4 (s = 2m²) bounds the number of extra relations only as "at most about m/2", which is not a proof.
- Parity at even s needs the d_2 formula for every k.
- **Status:** odd-s statements are Cat A at formula level. The even-s statements are dropped from the ledger.
- **Scope.** These are statements about closed forms. They equal the true d_cusp where dim V and d_Eis are proved: dim V = s(s−1) for odd s ≤ 51, together with the certified local factors.

1.2 Entry 17 (Gauss-sum vanishing; `stage_ps/.../P_s-Structure §A4`, `heavy-daily/2026-10-18-C_third-modularity/theoremG_check.py`)

Definitions.

- Q(n,k) = n² + 7nk + 7k(k+1)/2, which is integer-valued.
- For gcd(h,m) = 1, G(h/m) = (2m)^{−2} Σ_{n,k mod 2m} e(hQ(n,k)/m).

The summand is 2m-periodic in each variable, since Q(n+2m,k) − Q(n,k) and Q(n,k+2m) − Q(n,k) are both divisible by m.

Statement. G(h/m) = 0 iff 2 | m or 5 | m. Otherwise:

- |G| = 1/m if 7 ∤ m;
- |G| = √7/m if 7 | m.

The v1 ledger had "2 ‖ m". That is a transcription error: the source writes "2|m", an escaped "|".

Proof (verified line by line).

- **2 | m.** Q(n,k+m) − Q(n,k) = m(7n + 7k + 7(m+1)/2). With m even, the phase change is e(7h(m+1)/2) = −1, because 7h(m+1) is odd. The shift k ↦ k+m is a bijection mod 2m, so G = −G = 0.
- **m odd, 5 | m.** Put a = m/5. Then Q(n−a, k+a) − Q(n,k) = mn + a(7−m)/2. The phase is e(h(7−m)/10). Here (7−m)/2 ≡ 7·2^{−1} ≡ 1 mod 5 and gcd(h,5) = 1, so the phase is ≠ 1, and G = 0.
- **m odd, 5 ∤ m.**
- Q is m-periodic, so G = m^{−2} Σ_{n,k mod m}.
- With 2 invertible mod m, the substitution n′ = n + 7·2^{−1}k gives hQ ≡ h(n′² − (35/4)k² + (7/2)k).
- So G = m^{−2} g_1 g_2 with g_1 = Σ_{n′} e(hn′²/m), and $|g_1| = \sqrt{m}$.
- By CRT, g_2 factors over p^e ‖ m.
    - If p ∤ 35, the quadratic coefficient is a unit, and completing the square gives modulus

p^{e/2}.
    - If p = 7, the sum is Σ_{k mod 7^e} e(h′(−(5/4)k² + k/2)/7^{e−1}) = 7·Σ_{k mod 7^{e−1}}(...). Its modulus is 7·7^{(e−1)/2} = 7^{(e+1)/2}, because the quadratic coefficient is a unit mod 7.
- Hence |G| = m^{−2}·√m·√m·(√7 if 7 | m) ≠ 0.

Cross-check (Cat B, float). `theoremG_check.py` covers m ≤ 60 (1102 rationals) with 0 mismatches. The source reports m ≤ 105 (3374 pairs) with 0 mismatches.

1.3 Entry 23 (Hecke commutation; `SF/A7_s21/REPORT.md` A1)

Hypotheses.

- N ≥ 1, and p is prime with p ≡ 1 mod N.
- F is a smooth vector-valued function with F(γz) = ρ(γ)F(z) for γ ∈ $SL_2(Z)$, and F is Γ(N)-invariant (TB4).
- ΔF = λF.

Definition. T′_p F(z) = p^{−1/2}[F(pz) + Σ_{j=0}^{p−1} F((z+Nj)/p)].

Statement.

- T′_p F is of type ρ with the same λ, and it commutes with every constant matrix commuting with ρ (in particular the P_σ).
- If every component of F has the form Σ_{m∈(1/N)Z} a(m)√y K_0(2π|m|y)e(mx) + c√y, then T′_p F has coefficients b(m) = a(pm) + a(m/p), and constant term 2c.

Proof (verified). 1. Let Δ = {M ∈ $M_2(Z)$: det M = p, M ≡ I mod N}. The p+1 matrices (1, Nj; 0, p) and (p, 0; 0, 1) lie in Δ, because p ≡ 1 mod N. They are inequivalent under left multiplication by Γ(N), and since |Γ(N)\Δ| = p + 1 for p ∤ N, they form a complete system of representatives. 2. For γ ∈ $SL_2(Z)$, γ^{−1}Δγ = Δ, so {γ^{−1}Mγ} is again a system of representatives. 3. Then Σ_M F(Mγz) = Σ_M F(γ·γ^{−1}Mγ·z) = ρ(γ) Σ_{M′} F(M′z). Changing representatives does not change the sum, by Γ(N)-invariance. 4. Δ (the Laplacian) commutes with every z ↦ Mz for M ∈ $GL_2^+(R)$, so λ is preserved. The operator acts componentwise, so it commutes with constant matrices.

5. Fourier side. F(pz) contributes p^{1/2} a(m′/p) at frequency m′.

- Σ_j e(mNj/p) = p if p | mN and 0 otherwise. Since mN ∈ Z and gcd(p,N) = 1, p | mN iff m/p ∈ (1/N)Z.
- So Σ_j F((z+Nj)/p) contributes p·a(pm′)√(y/p) at frequency m′.
- After the factor p^{−1/2} this gives b = a(pm′) + a(m′/p). The √y-terms contribute c + c.

1.4 Entry 24 (sector lemma; `SF/A6_certify/REPORT.md` A1)

Hypotheses.

- TB2: u has type ρ_s and λ = 1/4. Its constant term is a multiple of c_0 = Π_W δ_0. The λ = 1/4 constant-term criterion holds.
- s is odd with 5 ≤ s ≤ 51, so dim V = dim W = s(s−1) and the components of u on W are linearly independent.

- σ is a P-even local-sign sector other than all-plus, with P_σW ≠ 0.

Proof (verified, with the hypothesis made explicit). 1. P_σ is a polynomial in the local reflections Z_p, which commute with ρ. So P_σu has type ρ_s and the same λ. 2. E = ρ-span(c_0) lies in the all-plus sector, because c_0 is even in every local component and the Z_p commute with ρ. Hence P_σc_0 = 0. The constant term of P_σu vanishes, so P_σu is cuspidal by the criterion. 3. If P_σu = 0, every row of P_σ would be a linear functional on W killing all values of u. By linear independence this forces P_σ|_W = 0, a contradiction. 4. Forms in distinct sectors are linearly independent, because the projectors are orthogonal.

The v1 hypothesis "P_σW ≠ 0" is equivalent to "P_σC ≠ 0" here, since P_σE = 0. **v1 omitted the range s ≤ 51; it is now explicit.**

2. Certificate and citation checks (re-runs in `A34/logs/`)

2.1 Certificate 10

- **Y = Z − SZS certificate.**
- The correct script is `SF/A8c_Spairing_even/scripts/pairing_on_W.py`; `SF/A8_Spairing_even/scripts/modp_trace.py` is not it. The script computes rank_$\mathfrak{p}$(Y_λ|_W) exactly mod $\mathfrak{p}$ on the orbit basis of W, for λ = (1) and (1,1), and checks $\hat{S}^2$ = |det| on W.
- Re-run (`logs/rerun_pairing_on_W.jsonl`): s = 8, 16, 18, 24, 26 give rank = dim W, identical to the source.
- Source values: full rank for s = 8, 16, 18, 24, 26, 32, 40, 48, 50, 56, 58. At s = 34 and 42 the rank is dim W − 2, so Y certifies nothing there. At s = 10, C = 0.
- **global_trE.py.** Re-run (`logs/global_trE_rerun_and_4_6mod8.txt`): s = 8, 16, 18 give identical results.

2.2 Certificate 56 (`SF/A8c_Spairing_even/scripts/galois_dilation_fast.py`)

What it computes. For every a ∈ (Z/L)^×, label dilation M_u and k with σ_a(T) = M_u T^k M_u^{−1} (exact monomial identity), it tests σ_a(Ŝ) = ±M_u Ŝ M_u^{−1} mod one prime $\mathfrak{p}$.

Why mod $\mathfrak{p}$ suffices. A char-0 identity survives reduction, and +1 and −1 cannot both hold mod $\mathfrak{p}$.

Results.

- Re-run at s = 8 and 10 (`logs/rerun_galois_dilation_8_10.jsonl`): T-compatible (a,u,k) counts 1024 and 4096, ε = +1 counts 128 and 256, ε = −1 count 0. Identical to the source.
- s = 16 and 18 were not re-run; the source log is `galois_dilation_fast_16_18.jsonl`.
- The controls s = 12 and 14 do have ε = −1 solutions, which exercises the code.

2.3 Certificate 57 (class-group predictors)

Predictor. As stated in `SF/A2/REPORT.md` B2, the predictor for the group X is dim S(ρ_s) = h(X)/2 if Cl(X) is cyclic, and h(X) if Cl(X) ≅ [2,2].

Exact class numbers. `scripts/classno_exact.py` counts cycles of reduced primitive indefinite forms. This is unconditional; PARI `quadclassunit` is GRH-conditional. Results are in `classno_exact.json`, and all

agree with `SF/A5_classgroup/classgroups_A5.json`.

s	d_K	h(K_s)	h(4n_s)	certified dim
9	28	1	2	1
11	44	1	4	2
23	1932	4	4	3
25	92	1	6	3
27	12	1	6	[3, 4]
29	348	2	6	[3, 4]

Exact conclusions (Cat A).

- **Cl(K_s).** The predicted value is h/2 for cyclic groups. It gives 1/2 at s = 9, 11, 25, 27 and 1 at s = 29, so it fails at all five.
- **Cl(4n_s).**
- At s = 23, h = 4, so the predicted value is 2 (if cyclic) or 4 (if [2,2]). Either way it is ≠ 3, so the predictor fails exactly at s = 23.
- At s = 25, Cl(4n_25) has order 6 and is therefore cyclic. The predicted value is 3 = dim S(ρ_25), so the predictor **holds** there. The v1 claim was wrong.
- At s = 27 and 29 the group is cyclic of order 6. The predicted value is 3, which lies inside the certified bracket, so this case is not decided (§3.3).

2.4 Certificate 58 (R-multiplicities on E at s = 56)

- The cited `exact_E0_R.py` certifies local classes only. The global (187, 186, 187) numbers come from `scan.py` (`logs_scan_s56.jsonl`), which is float64 SVD (Cat B).
- **Replacement proof (Cat A, assuming I1–I5).**
- **dim E ≥ 560.** This is a mod-ℓ Krylov rank, so a lower bound, from `SF/A31_entry7_check/global_trE.py` at two primes.
- **dim E is 504 or 560.** By the tensor factorisation (ledger entry 7(iii)), dim E = dim E0_{27} ·dim E0_7·dim E0_2. The odd factors are 14 and 4 (entry 7(i)). For the 2-adic classes (2,1/4) ⊕ (8, −u/16) of order 16, `SF/A11_induction/exact_E0.py` (`logs_exact_E0.txt`, exact_span_certified) gives dim 10 for u ≡ 1 mod 4 and 9 for u ≡ 3 mod 4.
- **So dim E = 560.**
- **Multiplicities.** At s = 56, (Exc) holds and s ≡ 2 mod 3. So by entry 11, tr_E R = −ω and tr_W R = −ω. Solving $n_1 + n_\omega + n_{\bar\omega} = 560$ and $n_1 + \omega n_\omega + \bar\omega n_{\bar\omega} = -\omega$ gives (187, 186, 187).
- **On C.** dim W = 812 = t(t+1) with t = 28 (orbit count). Then dim C = 252 and tr_C R = 0, so the multiplicities on C are (84, 84, 84).
- **Caveat.** This argument assumes the 2-adic component at s = 56 is one of the order-16 classes (2,1/4) ⊕ (8, −u/16) in A11's coverage list. The bound dim E ≥ 560 then excludes the dim-9 case.

2.5 Entry 2 (s ≡ 4, 6 mod 8): failed as stated, replaced

- `SF/A6_certify/REPORT.md` says the even-s analogue of step 2 (T^{2n} diagonal) "is computer-verified, not written for all s", for s = 6, 12, 14, 20, 22, 28, 30, 36, 38. So "for all s ≡ 4, 6 mod 8" is not proved.
- **New certificate.** `global_trE.py` was run for s = 12, 14, 20, 22, 28, 30, 36, 38, 44, 46. tr_E S = 0 at two primes each, an exact integer since |tr| ≤ dim E < ℓ/2.
- With tr_W S = 0 (entry 3), this gives tr_C S = 0 for every s ≡ 4, 6 mod 8 with 12 ≤ s ≤ 46. Here E = ρ-span(δ_0) in C[D], identified with the original E by entry 8.

3. Closure attempts

3.1 rev1 entry 15: distribution of r (odd s, formula level). Closed.

Notation.

- n = s(s−2) and k = ω(s) + ω(s−2).
- g(m) = $\prod_{q\|m} (1+1/q)/2$.
- 1 − r = 3·(s−2)/(s−1)·g(s)g(s−2).

All averages are over odd s ≤ X.

- **(a) Mean of k (elementary).**
- For odd p ≤ X, the number of odd s ≤ X with p | s(s−2) is X/p + O(1).
- Summing over p ≤ X: mean k = $\Sigma_{3\le p\le X} 2/p$ + O(1/log X) = 2 log log X + O(1), by Mertens.
- For the second moment, pairs p ≠ q contribute density 4/(pq) + O(1/X). Truncating at p, q ≤ X^{1/4} and bounding the tail by O(1) gives Var k ≪ log log X (Turán–Kubilius).
- **(b) The limit law of r is δ_1 (elementary).**
- By Chebyshev and (a), k ≥ log log X for all but O(X/log log X) odd s ≤ X.
- For those s, 1 − r ≤ 3(2/3)^k → 0.
- Since r ≤ 1, the empirical law of r converges to δ_1, and mean r → 1.
- **(c) Typical rate (elementary).** For almost all odd s:
- log(1−r) = −k log 2 + Y + O(1), where 0 ≤ Y ≤ $\Sigma_{p|n} 1/p \ll$ log log log s;
- and k = 2 log log s + O(ψ(s)√(log log s)) for any ψ → ∞.
- Hence 1 − r = (log s)^{−2 log 2 + o(1)}.
- **(d) Mean of g(s)g(s−2).**
- **Lower bound (elementary).** g ≥ 2^{−ω}, and x ↦ 2^{−x} is convex, so by Jensen the mean is ≥ 2^{−mean k}. That is ≫ (log X)^{−2 log 2}.
- **Upper bound ≪ 1/log X.** Use Nair–Tenenbaum, Acta Math. 180 (1998) 119–144, Theorem 1, with k = 2, $Q_1(n) = n$, $Q_2(n) = n-2$, and y = x.
    - Q = n(n−2) has no fixed prime divisor, since Q(1) = −1.
    - $F(n_1,n_2) = g(n_1)g(n_2)$, extended with g(2^e) = 1, satisfies $F(m_1n_1, m_2n_2) = g(m_1)g(m_2)F(n_1,n_2) \le F(n_1,n_2)$. So $F \in M_2(1, 1, \varepsilon)$.
    - The bound gives $\Sigma_{n\le X} F \ll X \prod_{p\le X}(1 - 2/p)(\Sigma_{n\le X} g(n)/n)^2$.
    - $\Sigma_{n\le X} g(n)/n \le \prod_{p\le X}(1 + 1/(2p) + O(p^{-2})) \asymp (\log X)^{1/2}$.
    - So the sum is ≪ X/log X. Restricting to odd n only decreases the sum, since F ≥ 0.
- **Consequence.** Mean(1−r) ≍ mean g between (log X)^{−1.386} and (log X)^{−1}. By Markov, the density of {r < 1−ε} is ≪ 1/(ε log X).

- **(e) Y has a continuous limit law.**
- *Existence.* Truncate at p ≤ y. By CRT, the truncated law is the law of Σ_{p≤y} X_p with independent X_p. The tail satisfies E|Y − Y_y| ≤ Σ_{p>y} 2/p² → 0.
- *Continuity.* Each X_p is discrete with largest atom P(X_p = 0) = 1 − 2/p for p ≥ 5, and ∏_p(1 − 2/p) = 0. By Lévy's theorem (P. Lévy, Studia Math. 3 (1931); see also Elliott, *Probabilistic Number Theory I*, Ch. 1), a convergent series of independent discrete variables has a continuous law iff the product of its largest atoms is 0.
- **Note.** This uses one classical cited theorem, stated with its hypothesis verified.
- **(f) Limit-point structure (elementary; source A5).** The set L of limit points of r satisfies L ⊆ {1} ∪ {1 − 3w : w ∈ W}, where W = {2^{−2−j}∏_{q∈S}(1+1/q)/2}.
- Along a subsequence with limit < 1, k is bounded. Pass to subsequences on which each prime-power factor is either constant or tends to ∞; each factor tending to ∞ contributes 1/2.
- So L is countable and closed, and L ⊆ {1/4} ∪ [1/2, 1].
- **Not included.**
- The Gaussian refinement (Halberstam) and the Chen-type disjunctions of A4 A6 are not needed.
- The Euler-product asymptotic B1 is Cat B.

3.2 Friction-coupled PDE (`SF/../maths/theorem-lemma-chain-9-maths.txt` Lemma 5 / Def. 1.1; `SF/A19_9maths_axioms` §1.6)

The source equation.

∂_t²Φ + γ_LQG F_friction(t)∇Φ + ρ_snap Θ_hyp(ρ_snap − ρ_DM) ∂_t²Φ = κ_dark sin(2πt/539.9) Φ + β_PBH ∂_t(e^{−M/k_BT} Φ).

- The full equation is **not well-formed**. γF∇Φ is a vector added to scalars, with no contraction specified, and F_friction is never defined. So "no periodic attractor" cannot be stated or proved for it. **Failed (not provable).**
- **Self-contained reduction (Cat A).** Take Φ = Φ(t), so ∇Φ = 0, and freeze c = ρ_snapΘ_hyp and g = e^{−M/k_BT} as constants. With ω = 2π/539.9 and A = 1 + c − βg, the equation becomes
- (★) AΦ″ = κ sin(ωt) Φ.
- **Theorem.** Assume κ ≠ 0.
- (i) For any Φ_0 and any (a,b) ≠ (0,0), the function Φ_0 + a cos ωt + b sin ωt does not solve (★). In particular the source's Φ_0 + 0.90 cos ωt + 0.18 sin ωt does not.
- (ii) If A ≠ 0, (★) has no solution that attracts all solutions.
- (iii) If A = 0, the only solution is Φ ≡ 0.
- **Proof.**
- (i) The left side contains only the frequency ω. The right side contains (κa/2) sin 2ωt − (κb/2) cos 2ωt. Both must vanish, so a = b = 0.
- (ii) (★) is the linear system x′ = M(t)x with x = (Φ, Φ′), where tr M = 0. By Liouville, the monodromy matrix has determinant 1.
    - Suppose a solution p attracts every solution, meaning |x(t) − p(t)| → 0 for all x. Since x − p ranges over all solutions, every solution tends to 0.
    - Then both Floquet multipliers have modulus < 1, contradicting det = 1.
- (iii) If A = 0, then κ sin(ωt)Φ = 0, so Φ = 0 wherever sin ωt ≠ 0, and by continuity Φ ≡ 0.
- The `checks.py` harmonic balance (sympy) agrees with (i).

3.3 Class-group predictor at s = 27 and 29. Failed to close.

- **Upper bounds.** s = 27 (3−,5−) ≤ 3 is certified (`SF/A9_hecke_family/certs/hcert_s27_3m_5m_M6_Q40.json`, re-read, not re-run). The (+,+) sector certificates for s = 27 and 29, and the s = 29 (−,−) certificate, never completed: OOM or box restart (`SF/A9/logs/queue_rerun.out`). I did not re-run them, because each needs hours of certifier time near the 3 GB limit.
- **Why upper bounds would not settle it anyway.** The Cl(4n) predictor value is 3. Refuting it needs **dim = 4, i.e. a lower bound of 4**, not an upper bound.
- **Attempt.** I copied `certify_hecke_s.py` to `A34/scripts` (unchanged) and ran it with 8 Hecke primes instead of 3 (`logs/hecke_27_29_NP8.out`, `certs/hecke_lower_s{27,29}.json`; 4 min each, < 0.7 GB).
- The rank of {P_σu, T′_p P_σu} stays at 2 in (−,−) and 1 in (+,+), at both s.
- So the Hecke span of the lift is exhausted. A third (−,−) form would need a different construction, such as a theta lift from another genus class.
- **Result.** dim S(ρ_s) ∈ [3, 4] at s = 27 and 29 (the upper bound 4 at s = 27 is not certified either, since (+,+) did not finish). Whether the Cl(4n) predictor fails there is undecided. The Cl(K_s) predictor fails exactly at both (§2.3).

3.4 E_L ↔ u_L (`SF/RTA_implications` §2). Not proved; left out.

- The claim identifies the ν ↦ −ν even part of the cone-sum series C_third,s with the λ = 1/4 Maass theta lift. It is Cat B by citation in its source (C_third-Unified).
- A proof would need the analytic comparison of the two generating functions: the indefinite theta integral against the cone sums, including the wall terms. That cannot be done within this check.
- No ledger entry uses it. Entry 25 (irreducibility of C at s = 7) is independent of it.

4. Other notes

- **Entry 1.** The proof is now the full A6 argument. It needs TB3 ($S^2 = P$ on the label space, i.e. $\rho(-I) = P$), which is added as a standing hypothesis.
- **Cat B kept out of the ledger.** All float scans, the A5 dims, and `theoremG_check.py` (float).

Files (all new)

- REPORT.md, REPORT.html, REPORT.pdf
- `scripts/classno_exact.py`, `classno_exact.json`
- `scripts/certify_hecke_s.py`, `coeffsS.py`, `liftS.py` (copies of A9's, unchanged); `certs/hecke_lower_s27.json`, `hecke_lower_s29.json`
- `logs/`: hecke_27_29_NP8.out, rerun_galois_dilation_8_10.jsonl, rerun_pairing_on_W.jsonl, global_trE_rerun_and_4_6mod8.txt